

FSOC Coarse Pointing, Acquisition & Tracking - Performance Report

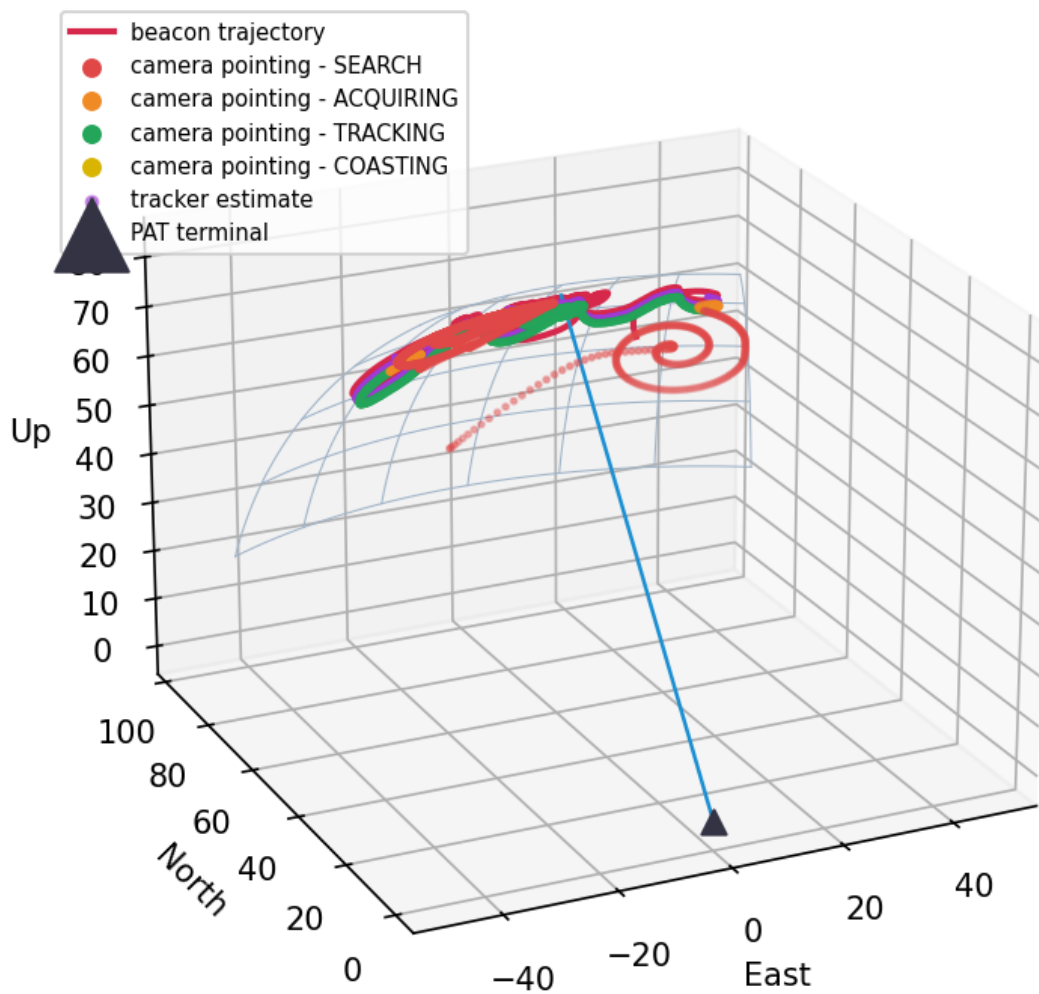
Run	
Generated	2026-10-02T15:40:39
Seed	467667441
Identification method	hybrid
Beacon path	random_walk at 2.48 deg/s
Decoys	4 (1 blinking)
Turbulence / vibration	1.51 / 0.49
Clouds	yes
Reported-position error (GPS prior)	4.73 deg
Camera start	az -4.2, el -12.0 deg (28.8 deg from the reported position)

Performance summary

Metric	Value
Simulation duration	120.0 s
Frames simulated	3600
Achieved frame rate	6.8 FPS
Processing time (mean / max)	9.81 / 119.67 ms
Processing throughput	102 FPS
Acquisition time	9.10 s
Time locked on beacon (whole run)	52.6 %
Lock retention (after first lock)	56.9 %
Lock retention while beacon visible	58.3 %
Lock losses	56
Tracking error mean / RMS / max	8.16 / 9.94 / 35.43 px
Tracking error mean / max	1.781 / 7.730 mrad
Ready for fine-pointing handover	71.3 % of locked time
False-lock frames	100
Candidates rejected / locks dropped	3 / 0
Beacon in field of view / occluded	59.4 / 2.3 %

3D view of the run

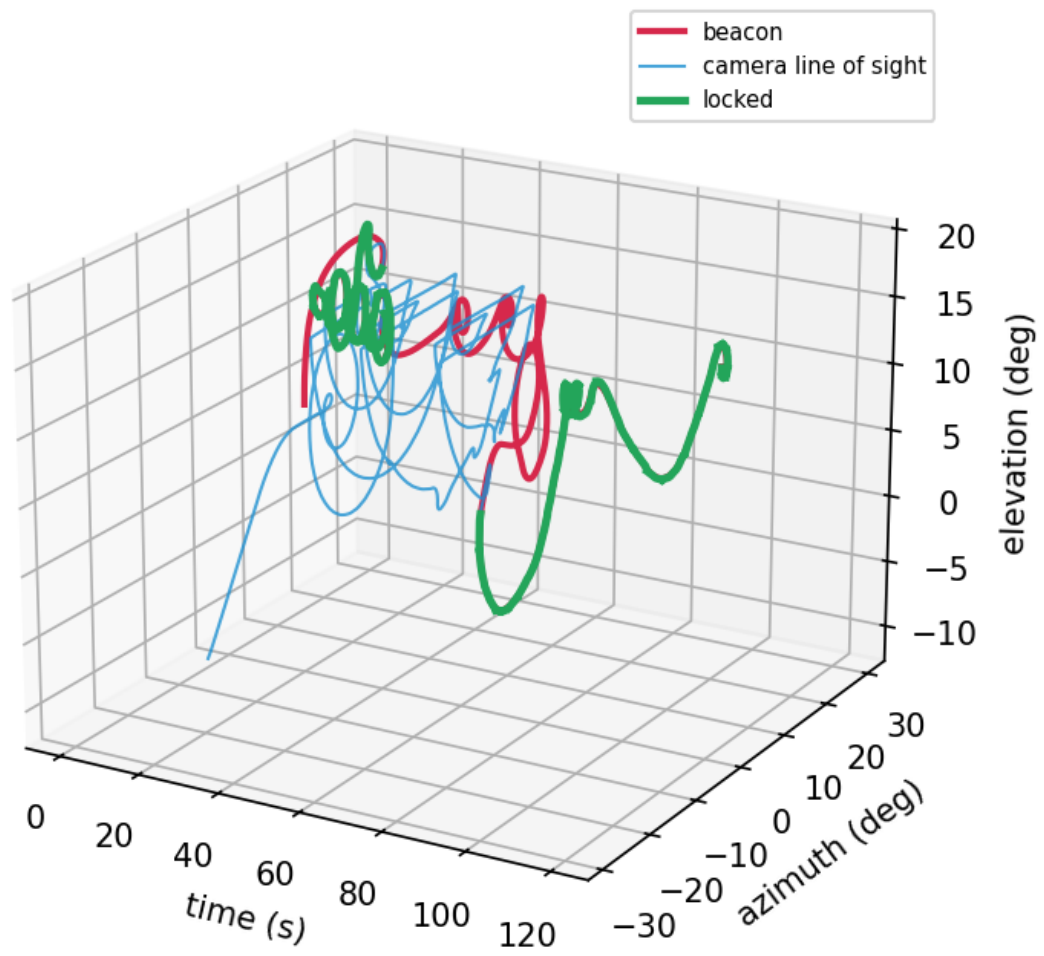
3D view: beacon trajectory and camera line of sight on the sky dome



Red: true beacon path. Dots: where the camera pointed, coloured by tracker state (red search, orange acquiring, green tracking, yellow coasting). Purple: tracker estimate.

3D view: azimuth / elevation over time

3D view: azimuth / elevation over time - beacon vs camera pointing



The camera line of sight (blue, green while locked) converging onto and following the beacon (red) shows acquisition and tracking in one picture.

Time charts

Tracking performance over time

